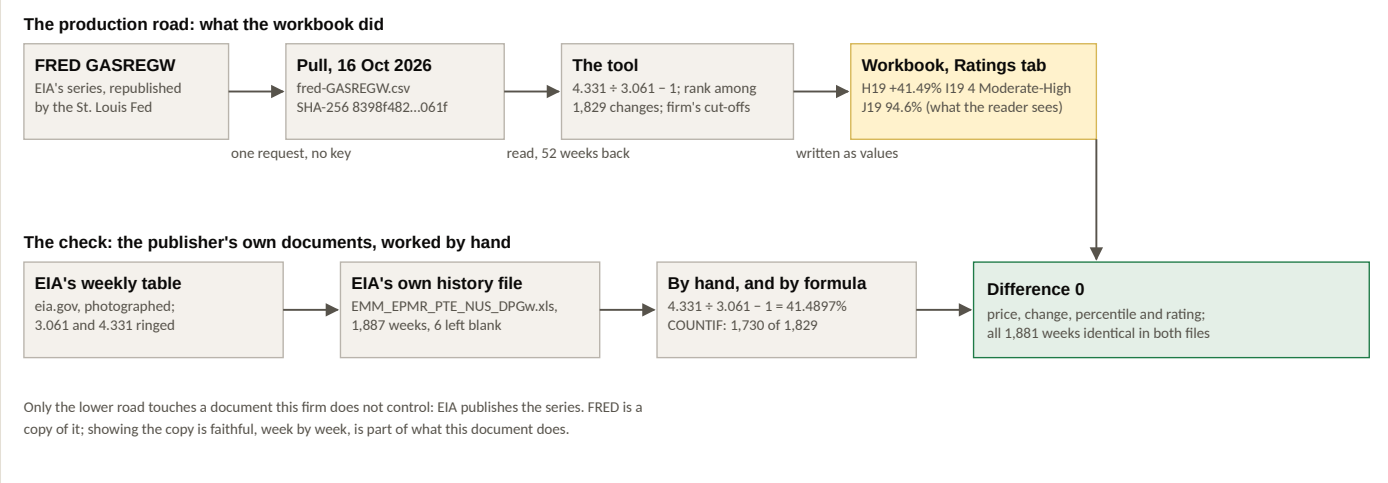
Tie-out: gasoline's change over a year, +41.49%, rated 4

The economic conditions workbook, Ratings tab, built 2026-10-16 05:17 UTC from the pull of 2026-10-16. The top reading on the tab, traced from the cell the reader sees to the publisher, the U.S. Energy Information Administration, and worked again by hand. Three more of the tab's headline figures are tied in the roster.

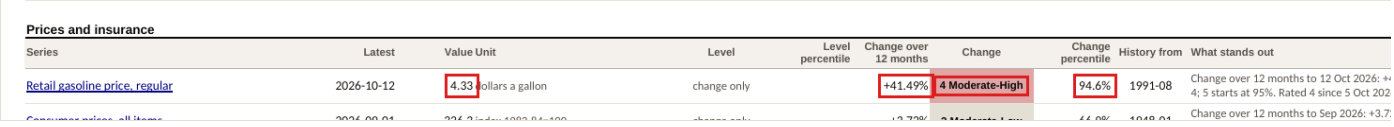
4 of 4 figures tied out to their publishers



1 · The figure

Where the reader sees it: **Ratings** tab, row 19, *Retail gasoline price, regular*, under *Prices and insurance*. Read out of the workbook file (`economic-conditions.xlsx`, SHA-256 `41b8edf53387b62e…`) with openpyxl, cell by cell, and photographed as LibreOffice shows it:

Cell	Stored value	Shown as	Meaning
H19	0.414897092453	+41.49%	change over 12 months to 12 Oct 2026
I19	4 Moderate-High	4 Moderate-High	its rating
J19	0.9458720612356478	94.6%	its percentile in its own history
D19	4.331	4.33	latest price, dollars a gallon (shown to two places)
Gasoline!O1852, O1904	3.061, 4.331	3.061, 4.331	the prices for the weeks of 13 Oct 2025 and 12 Oct 2026, on the series' own tab



Ours: the Ratings tab's gasoline row, the four cells ringed (workbook rendered in LibreOffice)

2 · The call

The pull made one request to FRED, with no key and no User-Agent (FRED stalls on one that names a browser):

```
curl -sS "https://fred.stlouisfed.org/graph/fredgraph.csv?id=GASREGW"
```

The response, saved as `work/macro-cache/2026-10-16/fred-GASREGW.csv`; the lines that carry the figure:

```
observation_date,GASREGW
2025-10-13,3.061
2026-10-05,4.354
2026-10-12,4.331
```

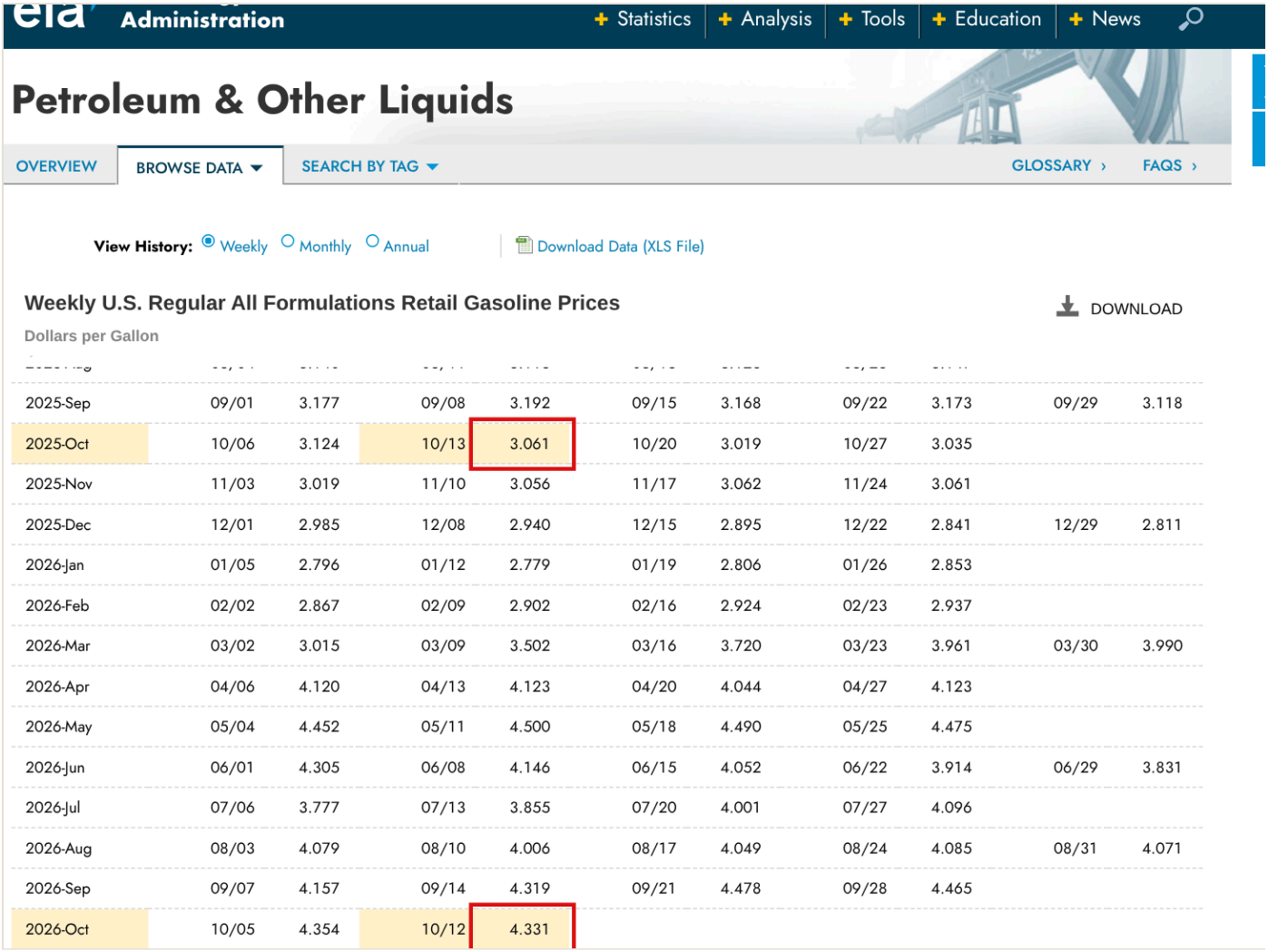
Its SHA-256, `8398f482c6043d6b02c988a268e57024d4427c46afce639be665a746488e061f`, is the fingerprint the workbook's **Sources** tab gives for that file, so this is the file the workbook read. It is kept beside this document as `pull-2026-10-16-fred-GASREGW.csv`.

3 · The derivation

1. Every week with a price is kept; a week FRED prints as `.` is left empty, never zero (1,881 priced weeks, 20 Aug 1990 to 12 Oct 2026).
2. For each week, the price 52 weeks earlier: the reading nearest the date 364 days before, at most three days away. For the week of 12 Oct 2026 that is the week of 13 Oct 2025, exactly 364 days.
3. The change: 4.331 ÷ 3.061 − 1 = **41.4897%**.
4. Its percentile: the share of all 1,829 such changes, 19 Aug 1991 on, that are at or below it (for gasoline a rise is worse): 1,730 ÷ 1,829 = **94.587%**.
5. Its rating: the firm's cut-offs (`docs/macro/settings.json`, 16 Oct 2026) start 2, 3, 4 and 5 at the 50th, 75th, 90th and 95th percentiles, and a reading on a line takes the worse rating. 94.587 is at or past 90 and short of 95: **4 Moderate-High**.

4 · The independent source

Publisher	U.S. Energy Information Administration (EIA)
Series	Weekly U.S. Regular All Formulations Retail Gasoline Prices, `EMM_EPMR_PTE_NUS_DPG`
Units	Dollars per gallon, as published (three places)
Where	eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=EMM_EPMR_PTE_NUS_DPG&f=W, *View history: Weekly*
Rows, columns	row **2025-Oct**, week **10/13**: **3.061**; row **2026-Oct**, week **10/12**: **4.331**
Fetched	16 Oct 2026, 05:27 UTC, in Chromium, certificate checks on



2025-Sep	09/01	3.177	09/08	3.192	09/15	3.168	09/22	3.173	09/29	3.118
2025-Oct	10/06	3.124	10/13	3.061	10/20	3.019	10/27	3.035		
2025-Nov	11/03	3.019	11/10	3.056	11/17	3.062	11/24	3.061		
2025-Dec	12/01	2.985	12/08	2.940	12/15	2.895	12/22	2.841	12/29	2.811
2026-Jan	01/05	2.796	01/12	2.779	01/19	2.806	01/26	2.853		
2026-Feb	02/02	2.867	02/09	2.902	02/16	2.924	02/23	2.937		
2026-Mar	03/02	3.015	03/09	3.502	03/16	3.720	03/23	3.961	03/30	3.990
2026-Apr	04/06	4.120	04/13	4.123	04/20	4.044	04/27	4.123		
2026-May	05/04	4.452	05/11	4.500	05/18	4.490	05/25	4.475		
2026-Jun	06/01	4.305	06/08	4.146	06/15	4.052	06/22	3.914	06/29	3.831
2026-Jul	07/06	3.777	07/13	3.855	07/20	4.001	07/27	4.096		
2026-Aug	08/03	4.079	08/10	4.006	08/17	4.049	08/24	4.085	08/31	4.071
2026-Sep	09/07	4.157	09/14	4.319	09/21	4.478	09/28	4.465		
2026-Oct	10/05	4.354	10/12	4.331						

Source: EIA's own weekly table, the page's title and units above, the two weeks ringed

Read twice. The same page's *Download Data (XLS File)*, `https://www.eia.gov/dnav/pet/hist_xls/EMM_EPMR_PTE_NUS_DPGw.xls` (SHA-256 `668bee46…233c`), sheet *Data 1*, carries the whole history: row 1838, 10/13/2025, 3.061; row 1890, 10/12/2026, 4.331. It has 1,887 rows, a row for every week, 6 of them blank (December 1990 to January 1991, when EIA published no price). It is kept beside this document as `eia-EMM_EPMR_PTE_NUS_DPGw.csv`, the sheet saved as CSV by LibreOffice.

5 · The comparison

	Ours: the workbook	Source: EIA	Difference
Price, week of 12 Oct 2026	4.331 (Ratings!D19; Gasoline!O1904)	4.331 (table; XLS row 1890)	0
Price, week of 13 Oct 2025	3.061 (Gasoline!O1852)	3.061 (table; XLS row 1838)	0
Change over 12 months	41.4897092453% (H19)	41.4897092453% (by hand)	0.0000000000
Percentile in its own history	94.5872061236% (J19)	94.5872061236% (1,730 of 1,829)	0
Rating, the firm's cut-offs	4 Moderate-High (I19)	4 (94.587 ≥ 90, < 95)	0

The change agrees to ten decimal places, which is as many as the tool keeps. **Week by week:** the FRED file the workbook read and EIA's own file hold the same 1,881 priced weeks, and not one price differs (1,881 weeks in both, 0 only in FRED, 0 only in EIA). So the road through FRED changed nothing for this series.

The four sameness checks. Same series: FRED's GASREGW is EIA's EMM_EPMR_PTE_NUS_DPG, all 1,881 weeks equal. Same dates: the weeks of 13 Oct 2025 and 12 Oct 2026. Same basis: the weekly retail price, taxes in, not seasonally adjusted, as EIA publishes it. Same units: dollars a gallon, three places (the workbook shows two and stores three). Verdict: **TIED**.

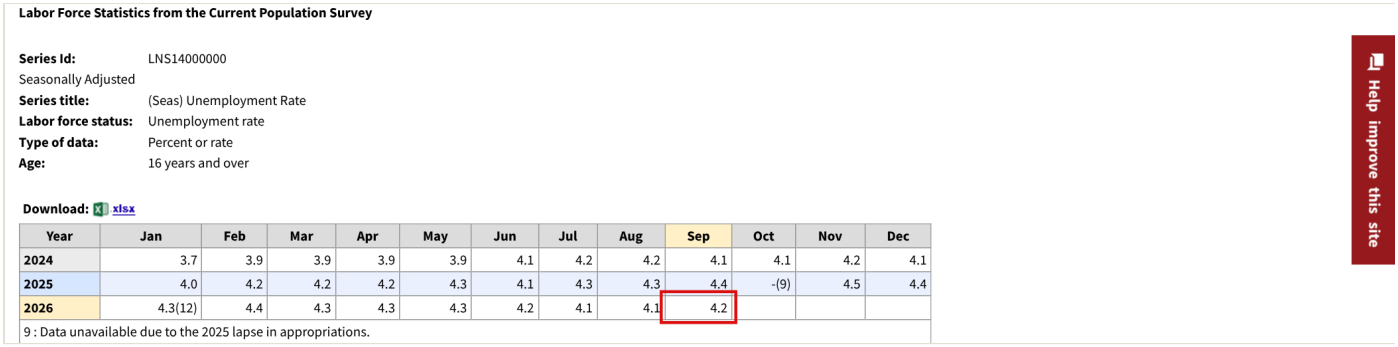
The roster

Verdict	Figures	Which
DIFFERS	0	
COULD NOT	0	
TIED	4	gasoline's change; the unemployment rate; the Sahm reading; government-backed mortgages 90+ days late, their change

Figure, as the Ratings tab shows it	Ours (the workbook)	Source	Diff	Verdict
Gasoline, change over 12 months to 12 Oct 2026: +41.49%, rated 4	H19 0.414897092453; I19; J19	EIA, above	0	**TIED**
Unemployment rate, September 2026	Ratings, D column: 4.2	BLS LNS14000000, 2026, Sep: 4.2	0	**TIED**
Sahm reading, September 2026	Ratings!C6: 0.00 in Sep 2026	FRED SAHMCURRENT, 2026-09-01: 0.00	0	**TIED**
Government-backed mortgages 90+ days late, change over 12 months to 2026 Q2	Ratings, D: 2.8; H: 0.9 points	FHFA NMDB: 2025Q2 1.9, 2026Q2 2.8; change 0.9	0	**TIED**

The unemployment rate: BLS's own table

The tool reads FRED's UNRATE, which is BLS series LNS14000000. BLS's data page, fetched 16 Oct 2026 in Chromium: series id, seasonally adjusted, ages 16 and over; 2026, September: 4.2. October 2025 shows `-(9)`, BLS's mark for the lapse in appropriations, which the tool keeps as an empty month.



Labor Force Statistics from the Current Population Survey

Series Id: LNS14000000
Seasonally Adjusted
Series title: (Seas) Unemployment Rate
Labor force status: Unemployment rate
Type of data: Percent or rate
Age: 16 years and over

Download: xlsx

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2024	3.7	3.9	3.9	3.9	3.9	4.1	4.2	4.2	4.1	4.1	4.2	4.1
2025	4.0	4.2	4.2	4.2	4.3	4.1	4.3	4.3	4.4	-(9)	4.5	4.4
2026	4.3(12)	4.4	4.3	4.3	4.3	4.2	4.1	4.1	4.2			

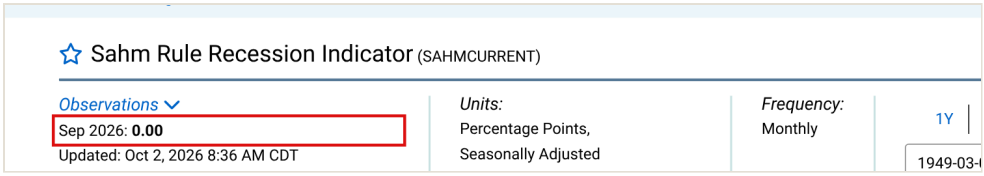
9 : Data unavailable due to the 2025 lapse in appropriations.

Help improve this site

Source: BLS, LNS14000000, 2026 September ringed

The Sahm reading: the St. Louis Fed's own

The tool works out its own Sahm reading from the unemployment rate; FRED publishes SAHMCURRENT, worked out by the St. Louis Fed. Two computations, one published: September 2026, 0.00 in both (and all 931 months agreed on 15 Oct 2026).



Source: FRED, SAHMCURRENT, the September 2026 observation ringed

Government-backed mortgages 90+ days late: FHFA's own file

Downloaded fresh from FHFA (SHA-256 `2083c4e1…4e89`, the same as the file the pull read). The rows, every column as filed:

Federal Housing Finance Agency, National Mortgage Database
fhfa.gov/document/d/nmdb/nmdb-mortgage-performance-statistics-national-census-areas-quarterly.zip → nmdb-mortgage-performance-statistics-national-census-areas-quarterly.csv
SHA-256 of the zip: 2083c4e1b5273fd2f471d2e58057b17b82b86ef3ac7ed13c58925b4453ec4e89 (fetched 16 Oct 2026). Every column of the five rows, as filed; P90DL = percent of mortgages 90 or more days late.

GEOLEVEL	GEOID	GEONAME	MARKET	PERIOD	YEAR	QUARTER	SUPPRESSED	SERIESID	VALUE1
National	USA	United States	Government / Non-Conventional	2025Q2	2025	2	0	P90DL	**1.9**
National	USA	United States	Government / Non-Conventional	2025Q3	2025	3	0	P90DL	2.1
National	USA	United States	Government / Non-Conventional	2025Q4	2025	4	0	P90DL	2.6
National	USA	United States	Government / Non-Conventional	2026Q1	2026	1	0	P90DL	3
National	USA	United States	Government / Non-Conventional	2026Q2	2026	2	0	P90DL	**2.8**

Source: FHFA, National Mortgage Database file, the two quarters ringed

How to run it yourself

1. Open `https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=EMM_EPMR_PTE_NUS_DPG&f=W`. In row **2025-Oct**, the price beside **10/13** is **3.061**; in row **2026-Oct**, beside **10/12**, **4.331**. Find them by row and week: 3.061 also appears beside 12/03 in 2007 and 11/24 in 2025.
2. Divide: `4.331 / 3.061 - 1` = **0.414897**, which is +41.49%.
3. Click *Download Data (XLS File)* and open sheet *Data 1*. In cell `C56` type `=IF(AND(ISNUMBER(B4),ISNUMBER(B56)),B56/B4-1,"")` and fill it down to `C1890` (each week against the row 52 above: the sheet has a row for every week). Then `=COUNTIF(C56:C1890,"<="&C1890)` gives **1730**, `=COUNT(C56:C1890)` gives **1829**, and their ratio **0.9459** (run in LibreOffice on 16 Oct 2026: 0.945872061235648).
4. 94.59% is at or past 90 and short of 95, the firm's cut-offs for 4 and 5: **4 Moderate-High**.
5. Open the workbook's **Ratings** tab: row *Retail gasoline price, regular* shows +41.49%, 4 Moderate-High, 94.6%.
6. Or run the check. From `threshold-engine/`, make the workbook as the procedure does, then fetch the three roster sources into this folder and run it (the EIA file is already here):

```
python -m threshold_engine.macro pull --cache work/macro-cache
python -m threshold_engine.macro rate --cache work/macro-cache --settings
docs/macro/settings.json --workbook work/economic-conditions.xlsx
cd docs/macro/tie-out/gasoline-change-2026-10-16
curl -sS -A "Mozilla/5.0" -o bls-LNS14000000.html
"https://data.bls.gov/timeseries/LNS14000000?
from_year=2024&to_year=2026&output_view=data"
curl -sS -o fred-SAHMCURRENT.csv "https://fred.stlouisfed.org/graph/fredgraph.csv?
id=SAHMCURRENT"
curl -sS -A "Mozilla/5.0" -o fhfa-nmdb.zip "https://www.fhfa.gov/document/d/nmdb/nmdb-
mortgage-performance-statistics-national-census-areas-quarterly.zip"
python check.py ../../../../work/economic-conditions.xlsx ../../../../work/macro-
cache/2026-10-16 eia-EMM_EPMR_PTE_NUS_DPGw.csv bls-LNS14000000.html fred-SAHMCURRENT.csv
fhfa-nmdb.zip ../../settings.json
```

Use the dated pull folder the workbook names on its Run tab.

7. On a later day the latest week moves on, and so do the numbers; the two weeks above stay where they are in EIA's table.

What it found

- **Every figure tied; nothing in the workbook needed correcting.** The price, the change, the percentile and the rating all came out the same from EIA's own documents, worked a second time without the tool's code.
- **The road through FRED costs nothing for gasoline:** all 1,881 priced weeks are identical in FRED's copy and EIA's own file, the gaps of 1990 and 1991 included.
- **A figure found by its text can be the wrong week.** EIA's table prints 3.061 in three weeks (3 Dec 2007, 13 Oct 2025, 24 Nov 2025). A reader checking by eye, or a tool matching on text, can land on the wrong one and still read 3.061. The how-to above finds it by row and week for that reason.
- **The spreadsheet route works on EIA's own file as published:** one formula, filled down, gives the same 1,730 of 1,829, so a reader with Excel and no Python can redo the rating.

What I got wrong

- **My first picture of EIA's page ringed the wrong week.** I placed the ring on the first cell reading 3.061, which is the week of 3 Dec 2007. The number was right and the week was wrong, and it would have passed this document's red-ink check. I saw it only when I cropped the picture and read the row label. The ring is now placed by row and week.
- **My first ring on FRED's page marked nothing.** The first element reading 0.00 was a hidden one at the page's corner. The picture now rings the visible *Sep 2026: 0.00*.
- **My first picture of the workbook missed the gasoline row.** The rings were drawn, but on the page outside the crop; the picture showed the rows beneath. Recropped from the row's position on the page.
- **Earlier in this work I said a fix was made when it was not:** a chart-label format, claimed fixed, did nothing in LibreOffice. The walk of the same day found it (W7 in the walk's defects).

What this does not prove

- **Four figures of 158 readings.** The other 154 rest on the same code, on the comparison of 12-month changes with FRED's own for 13 series (15 Oct 2026, every date), and on the Sahm comparison over 931 months; none of them is tied to its publisher.
- **FRED's copies of the other series are assumed faithful.** Shown here for gasoline only.
- **The FHFA figure is checked against the same file the tool read** (a fresh download, identical fingerprint). That proves the tool read FHFA's file correctly, not that the file is right. A second rendering, FHFA's online charts, was not tried.
- **The Sahm figure is two computations from one BLS series:** it proves the arithmetic, not the unemployment data, which the BLS row above ties separately.
- **Percentiles move when history is revised.** These are as published on 16 Oct 2026; whether EIA revises past weekly prices was not checked.
- **The workbook was opened in LibreOffice, not Excel.** Cell values were read from the file with openpyxl; how Excel displays them was not seen.
- **A rating is the firm's settings applied to a history.** It says where +41.49% ranks among gasoline's own changes, nothing about what it means for borrowers.